MIL-C-27278B(USAF)
5 July 1973
SUPERCEDING
MIL-C-27278A(USAF)
10 August 1967

MILITARY SPECIFICATION

CHECKLISTS, FLIGHT CREW

1. SCOPE

1.1 This specification covers detailed requirements for the first six (6) checklists covered by the general specification MIL-C-38778.

1.2 CLASSIFICATION

TYPE 1 FLIGHT CREW

Class 1 - Pocket-sized checklists for insertion in standard plastic insert.
Class 2 - Pocket-sized checklists for attachment to clipboard.
Class 3 - Scroll-type checklists
Class 4 - Cargo Loading checklists (MIL-C-9927)
Class 5 - Non-nuclear Munitions Delivery Checklists (MIL-M-38384)
Class 6 - Nuclear Munitions Delivery checklists (MIL-T-25832, MIL-T-25848) (MIL-B-27026, MIL-T-27579)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-38778 Checklists: Title Page, List of effective pages, Printing and [illegible] Binders; General Requirements for preparation of Flight Manuals

MIL-M-7700 Flight Manuals
(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Source of data. The data for each checklist shall be drawn directly from the applicable parent manual, specified by the procuring activity (see 6.2). (A parent manual is a manual which contains procedures from which the checklist is developed). The checklist for any given crewmember of an aircraft shall contain all the applicable material from the parent manuals.

3.1.1 Basic checklists. Any Class 1 or 2 checklist (see figures 1, 2, and 3) for any crewmember, which is developed from data derived from a flight manual prepared in accordance with MIL-M-7700, shall be the basic checklists and shall contain all the information required by the crewmember to function as a part of the crew complement.

3.1.2 Supplementary checklists.

3.1.2.1 Class 3 checklists. The Class 3 checklist (see figure 4) shall be developed from data taken from the normal procedure section of the parent (flight) manual. Line items may further be abbreviated to fit the items to one line. Otherwise, Class 3 checklists shall duplicate the information contained in the applicable portions of the Class 1 or 2 checklists. The Class 3 checklist shall contain only the normal procedures that can be accomplished while occupying the applicable crew station. If the normal procedure section of the parent manual contains a paragraph heading

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with a cross reference to another section of the manual for the procedure, the referenced procedure may be included in the Class 3 checklist. Under the present state of the art for the material used in the preparation of scrolls changes must be issued in the form of Revisions. For this reason, the scrolls shall show a Revision date in contrast to a basic date and change date.

3.1.2.2 Class 4, 5, or 6. Any Class 4, 5, or 6 checklist for any crewmember, which is developed from data derived from a parent manual not prepared in accordance with MIL-M-7700, shall be supplementary to the basic checklist for the crewmember. Class 4, 5, or 6 checklists shall be limited to specific functions or tasks. Supplementary Class 4, 5, or 6 checklists may be constructed to be used independently in whole or in part. They may be collatable in whole or in part into other related checklists. Or they may be integrated into a related checklist. If integrated checklists are desired, the primary line items in the applicable procedure section of the parent manual (usually the flight manual) shall be included in integrated checklists. Checklists that are to be collated shall be prepared so that procedure phases or information phases shall not be split by collation.

3.2 Materials. The Class 4, 5, and 6 checklists shall be printed on the same type of material used for the flight manual checklist (Class 1 or Class 2).

3.3 Applicability and compatibility. When specified by the procuring activity and when required to facilitate coordination between the flight and ground crewmembers in checking the aircraft, the Class 1 or 2 checklists shall be provided for the ground observer. All checklists for a crewmember of a given aircraft shall be compatible.

3.3.1 Crewmember checklist. Some aircraft may have multiple crewmembers with many common duties and few specialized duties. The procuring activity may in such cases authorize a "crewmember" checklist in which applicable crewmembers would be identified with appropriate procedures.

3.4 Concurrency of checklists with manuals. Checklists shall be issued simultaneously with the initial issue of the applicable parent manual. Checklists shall be modified simultaneously with a change or a revision to the applicable manual. Only checklists, however, that are affected by the change or the revision to the applicable parent manual shall be modified. As an example; a change or a revision to the parent manual affecting the pilot's checklist but not affecting the navigator's checklist shall result in modification of the pilot's checklist but not in modification of the navigator's checklist. The form of modification (change or revision) for the manuals and the checklists need not be identical. (Note: The preceding requirements could create situations where the issue and the change dates of the checklists and the applicable parent manual will not be identical.)

3.5 Format and arrangement. The checklists shall be prepared in accordance with MIL-C-38778, which shall govern in the event of a conflict.

3.5.1 Security classification. Every effort shall be made to issue the checklists as unclassified. Provided that the sense and the purpose of the instructions are not obscured and the classified information is not compromised, unclassified checklists.

3.5.2 Sequence. Unless the sequence has been set by precedent, specified herein, or specified by the procuring activity (see 6.2), the sections or parts of the checklist shall be in the same sequence as the sections or parts of the parent manual. Except for the emergency procedures in the Class 2 checklists may be in random order. Unless the procuring activity specifies otherwise (see 6.2), the takeoff and landing data (TOLD) cards shall be placed after the last page of the normal procedures of the checklist, the section containing the normal procedures of the checklists shall be arranged to preclude splitting any procedural phases by a TOLD card.

3.5.3 Page numbering. Each page of the introduction shall be numbered. The page numbers of the introduction shall be i, ii, etc. Each page of the sections or parts shall be numbered. If more than one version of a section or part is included in the parent manual, the page numbering of the section or part shall correspond to the number of versions in the parent manual. For example: If three versions of the performance section or part are included in the parent manual, the first page of the first,

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second, and third versions of the section or part in the checklist shall be numbered PA-1, PB-1, and PC-1, respectively. The length of the pages of the emergency procedures of the Class 2 checklist shall be staggered in increments of ¼ inch to permit visual indexing of the procedures at the bottom of the pages. Page E-1 shall be the first page of the emergency procedures of the Class 2 checklist. The last page of the emergency procedures of the Class 2 checklist shall be 7 inches long and shall be the longest page in the emergency procedures of the Class 2 checklist. Except for the emergency procedures in the Class 2 checklist, each section or part of the checklist shall begin at the top of a right-hand page.

3.5.4 Tab titles. Each emergency procedure page of the Class 2 checklist (see figure 3) shall have a tab title on each side of the page number. The tab title at the right of the page number shall identify the data on the page. The tab title at the left of the page number shall identify the data on the preceding page.

3.5.5 Standard terms and abbreviations. Standard terms and abbreviations used in the checklist shall conform to the requirements of Appendix B of MIL-M-7700. Line items appearing in the parent manual shall not be abbreviated in the checklist.

3.5.6 Splitting procedures. Certain procedures such as before takeoff, descent, and emergency procedures require the uninterrupted attention of crewmembers. Where practical each of these procedures shall be presented in such a manner that a crewmember can read the entire procedure without turning a page. To accomplish this as much as a a half a page may be left blank. This requirement need not be applied to procedures such as exterior inspection, interior inspection, and before leaving aircraft. If a procedure is continued on the next page, the word continued shall not be inserted at the bottom of the page.

3.6 Contents. Each checklist shall contain a title page, a list of effective pages, an introduction, and Sections or Parts. If the checklist contains abbreviations or if a separate applicability code is used in the checklist, this information shall be included in the introduction. If approved by the procuring activity, the checklist may contain unnumbered blank pages so that the crewmembers can notate desirable information.

3.6.1 Title page. The title page shall include the crew position of the crewmember to whom the checklist is applicable, the aircraft designation as specified in the applicable parent manual, the dates of the basic checklist and of the changes (if applicable), and any required additional information specified by the procuring activity (see 6.2).

3.6.2 List of effective pages. The list of effective pages shall contain the information specified by the procuring activity (see 6.2).

3.6.3 Introduction. The introduction shall contain a brief explanation of the purpose and of the scope of the checklist. The introduction may contain a brief explanation of the various parts of the checklist and any other relevant information associated with the use of the checklist. A deficiency report reflecting an error in the parent manual should not be submitted as a deficiency report on the checklist. The deficiency report submitted on the parent manual will suffice to correct the checklist. The information in the introduction of the checklist should not duplicate the information contained in the introduction of the parent manual.

3.6.4 Illustrations. Unless approved by the procuring activity, illustrations shall not be included in the checklists (see 6.2). When approved by the procuring activity, illustrations required to present necessary operating information may be extracted or derived from the parent manual to assure accurate reference data required for flight planning or mission execution. No figure numbers are required, and Titles should correspond to those used in the parent manual.

3.6.5 Section or parts

3.6.5.1 Table of contents. Each section or part of the checklist shall have a table of contents. The table of contents shall be the first entry following the title of the section or part. The table of contents shall list each primary heading contained in the section or part and the first page number on which it appears. When specified by the parent manual manager (see 6.2), important subordinate headings shall

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also be included in the table of contents. Care shall be taken so that the table of contents will not be too long; a single entry shall designate the general subject matter of a large number of closely related procedures or information grouped on one or two pages.

3.6.5.2 Introduction. An introduction to a section or part shall include any special instructions necessary for the proper use of that section or part.

3.6.5.3 Explanation of informational data. Each section or part containing informational data shall include an explanation of the derivation and the limitation of the informational data and of how the informational data is to be used. (As an example: Performance data for the checklist is derived by condensing performance data from the parent manual and shall not be used for mission planning or for determining TOLD card entries. It may be used for cross checking derived information, for rough estimations, or for recomputations of data when the parent manual is not attainable such as during flight in aircraft that are too space limited to accomodate the flight manuals.

3.6.5.4 Procedures. The sections or parts of the checklists shall contain procedures (i.e., normal and emergency), as applicable to the aircraft and the crewmembers. The emergency procedures in the checklist may be subdivided into ground operations, takeoff, inflight, and landing. The procedures of the checklist shall be derived by abbreviating the procedures and eliminating the amplifications of the procedures in the procedure sections of the parent manual applicable to the crewmember for whom the checklist is intended. Each primary line item that will affect the applicable crewmember shall be duplicated in the checklist in abbreviated format. (As an example, the primary line item "Wing Flap Lever - DOWN to 40% Neutral.") The inclusion of sub items in abbreviated format will be optional. Except for the most critical warnings and cautions, explanatory material shall not be included in the procedures of the checklist. A line item may be amplified in the parent manual be a warning or a caution. They shall be indicated by the addition of "W" for warning and "C" for caution after the line item in the checklist when the warning or caution is not included in the checklist.

3.6.5.5 Applicability code. If space permits, the applicability of a step that is restricted to a specific aircraft model, series, or equipment shall be coded in the checklist with the applicability code used in the parent manual. If space does not permit using the applicability code used in the parent manual, a separate applicability code will be permitted for the checklist and shall be listed in the introduction.

3.6.5.6 TOLD cards. All checklists prepared for crewmembers directly involved in takeoff and landing the aircraft shall contain TOLD cards (see figure 5). The TOLD card shall not contain page numbers. The procuring activity may require transparent plastic sheets which can be written upon or erased to insure that TOLD cards can be reused. If required by the procuring activity these sheets shall be placed before and after as necessary in Class 2 checklists. The TOLD cards shall contain takeoff conditions, takeoff data, landing immediately after takeoff data, landing at destination conditions, and landing at destination data. Either one or two TOLD cards may be used for each checklist. If two TOLD cards are used, the data for the TOLD cards shall be divided so that the takeoff conditions, takeoff data, and landing immediately after takeoff data are on one card and the landing at destination conditions and landing at destination data are on the other card. The procuring activity will determine the form to be used. The following items are typical of items contained in each phase of the TOLD cards.

(1) Conditions at Departure and Destinations

Runway Lehgth_____ Runway Height_____ft

Runway Gradient_____ Runway Surface Condition_____

Gross Weight GW_____LBS

Equivalent Gross Weight EGW_____LBS

Center of Gravity (CG) _____%MAC

Cross Wind Component X wind_____knts

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Head Wind Component Headwind_____knts

Outside Air Temperature OAT_____°C

Pressure Altitude PA_____FT

Dew Point DPT_____°C or °F

Runway Condition Reading RCR_____

Crosswind Gear Setting Xwind_____°or Units

TAKEOFF DATA

Engine Pressure Ratio EPR_____

Min Torque Pressure BMEP_____PSI

Carburetor Air Temp CAT_____°C

Flap Setting_____° or units

Refusal Speed/Distance V_R_____Knts Dis_____Ft

Rotation Speed V_{rot}_____Knots

Engine Out Climb Speed _____Knots

LANDING

Immediately After Takeoff Destination

Approach Speed V_{APP}_____Knots

Flare Speed V_{TD}_____Knots

Touchdown Speed _____ Knots

Ground Run Grnd Run Max_____Ft Min_____Ft

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be as specified in MIL-M-7700 and MIL-M-38778

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-38778.

5.2 Folding and fastening. Class 3 checklists shall not be folded.

6. NOTES

6.1 Intended Usage. The checklists covered by this specification are intended to be used as aircraft operating procedures by flight crews.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type and class of checklist (see 1.2 and 1.3)

c. Applicable Parent Manual. (See 1.2 and 3.1)

d. Applicability to crewmembers (See 3.3)

e. Information required for sequence (See 3.3.3)

f. Information required for placing of (TOLD) cards (See 3.3.2)

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g. Requirements for blank pages (See 3.6))

h. Information required for the cover page (See 3.6.1)

i. Information required for list of effective pages (See 3.6.2)

j. Procedures for submission of deficiency reports (See 3.6.3)

k. Requirements for table of contents (See 3.6.5.1)

6.3 Responsible activity. The activity responsible for the checklist is the same as that for the flight manual of the affected aircraft.

6.4 Review of reproducible copy. Review of reproducible copy by Government activities is not intended to reduce or replace the contractor's responsibility for assurance of technical accuracy.

6.5 Safeguarding classified information. Instructions regarding the safeguarding of classified information are contained in DOD 5220.22-M Industrial Security Manual. Copies of this manual may be obtained from the procuring activity or as directed by the contracting officer.

6.6 Class of checklists to be ordered. Unless prior approval for the Class 2 or 3 checklist has been obtained from the flight manual manager, the Class 1 checklist will be ordered. The Class 1 checklist is preferred in the interest of standardization and cost reduction.

Custodian:

Air Force - 11

Review Activities:

Air Force 10, 70, 71, 80, 82, 84

Preparing Activity

Air Force - 16

Project TMSS F368

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T.O. 1F-111D-1CL-1

PREFLIGHT CHECK.

BEFORE ENTERING COCKPIT. (AC)

1. Eject handle safety pins (2)—Installed.
2. Canopy center beam safety pins (3)—Installed.
3. Left canopy detach handle safety pin—Installed. (If applicable).
4. Quick rescue kit—Stowed (If applicable).
5. Emer oxy press—Check.
6. Aft console—Checked:
   a. All circuit breakers—In.
   b. Computer power switches—ON.
   c. CADC power sw—POWER.
   d. Ground ignition cutoff sw—NORM.
   e. Gyros power sw—GYROS.
   f. R&L cowl switches—NORM.
   g. Mach trim test sw—NORM.
   h. Fire detect switches—NORM.
7. Utility light—Stowed as desired.

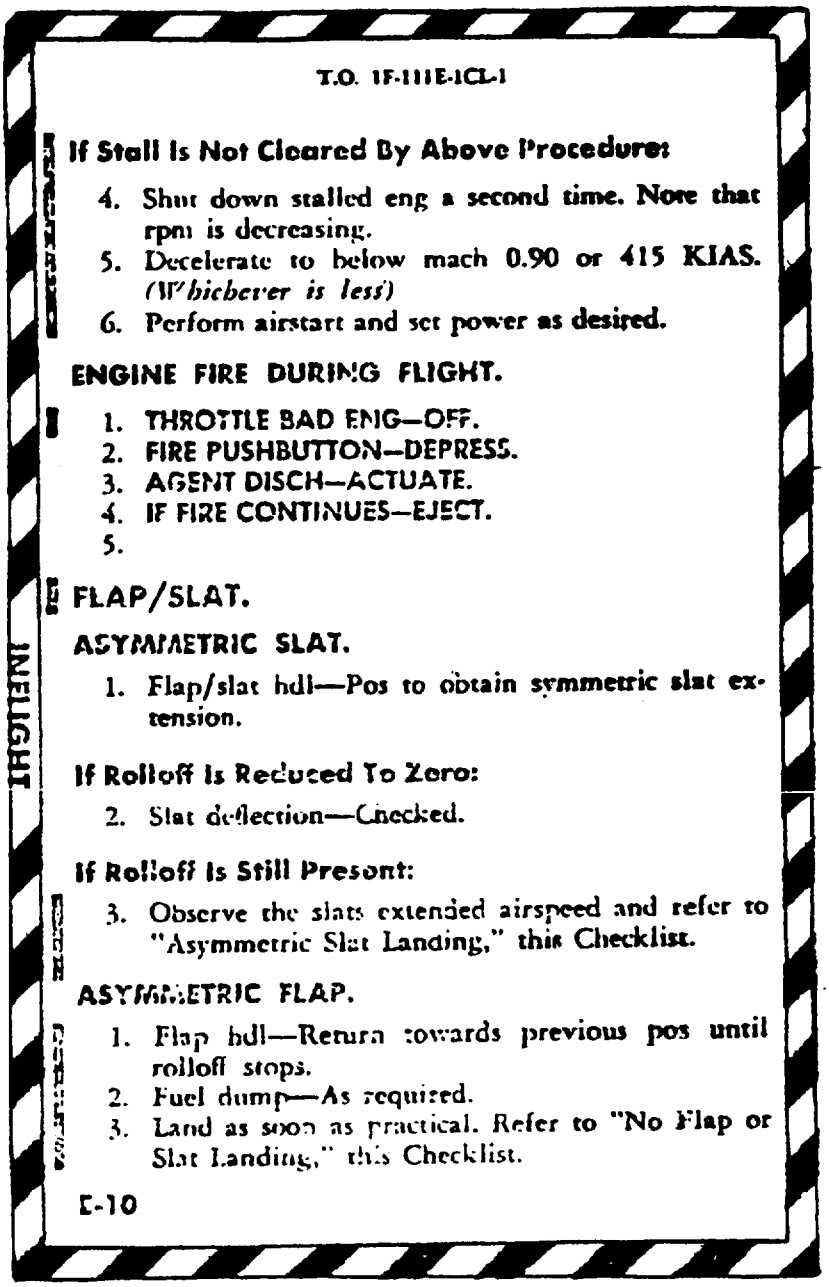
BEFORE ENTERING COCKPIT. (WSO)

1. Canopy external emergency release handle—Check (If installed).
2. Ejection handle safety pins (2)—Installed.
3. Canopy center beam safety pins (3)—Installed.
4. Canopy ext emer initiator pin (1)—Check installed (If applicable).
5. Survival equip compartment door—Checked.
6. Crew mod chaff disp cont lever—As required.

N-2

Figure 1. Example of Page of Class 1 or 2 Checklists

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T.O. 1F-111E-1CL-1

If Stall Is Not Cleared By Above Procedure:

4. Shut down stalled eng a second time. Note that rpm is decreasing.
5. Decelerate to below mach 0.90 or 415 KIAS. *(Whichever is less)*
6. Perform airstart and set power as desired.

ENGINE FIRE DURING FLIGHT.

1. THROTTLE BAD ENG—OFF.
2. FIRE PUSHBUTTON—DEPRESS.
3. AGENT DISCH—ACTUATE.
4. IF FIRE CONTINUES—EJECT.
5.

FLAP/SLAT.

ASYMMETRIC SLAT.

1. Flap/slat hdl—Pos to obtain symmetric slat extension.

If Rolloff Is Reduced To Zero:

2. Slat deflection—Checked.

If Rolloff Is Still Present:

3. Observe the slats extended airspeed and refer to "Asymmetric Slat Landing," this Checklist.

ASYMMETRIC FLAP.

1. Flap hdl—Return towards previous pos until rolloff stops.
2. Fuel dump—As required.
3. Land as soon as practical. Refer to "No Flap or Slat Landing," this Checklist.

E-10

Figure 2. Example of Emergency Procedure Page (Class 1 Checklist)

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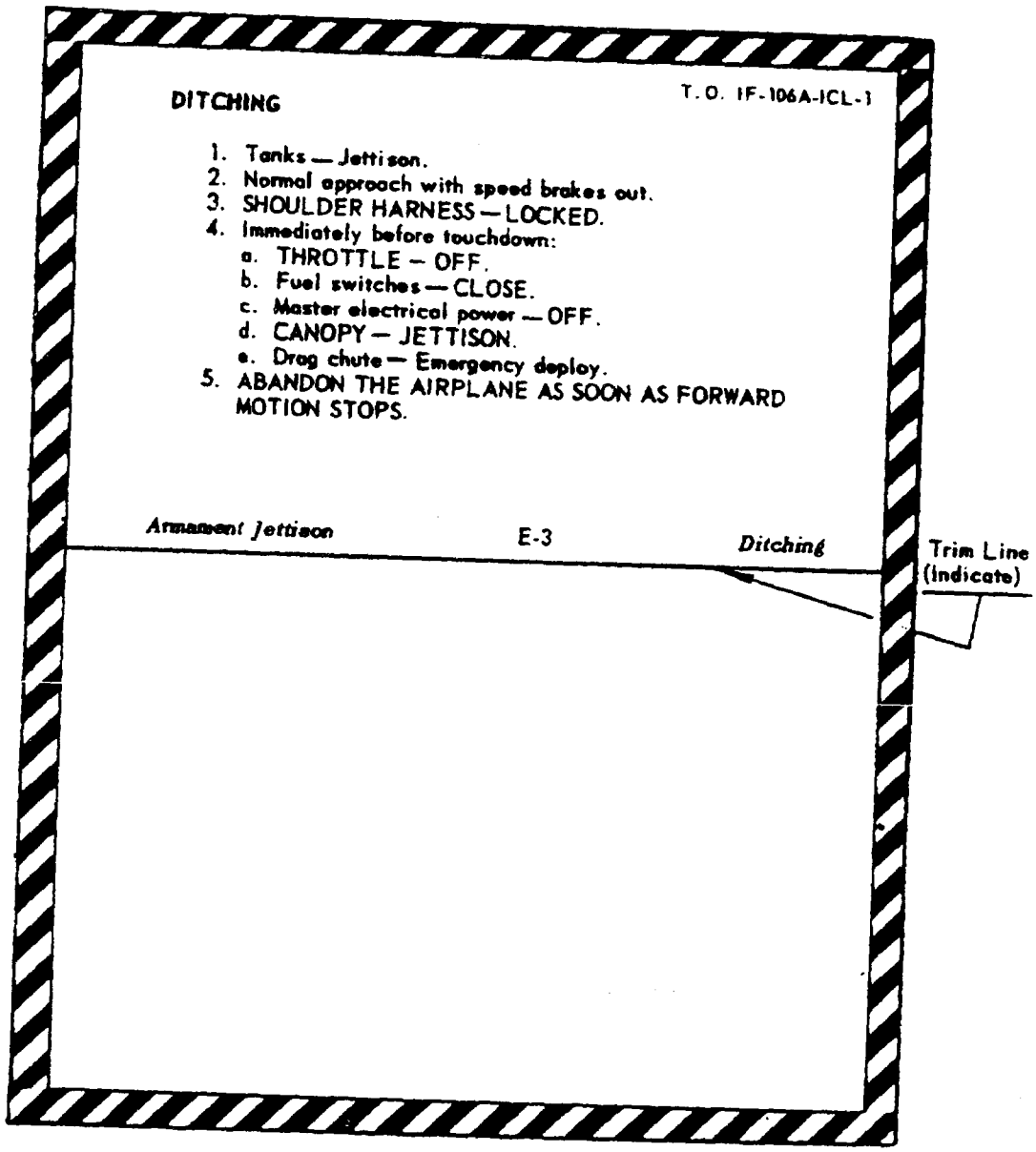


Figure 3. Example of Emergency Procedure Page (Class 2 Checklist)

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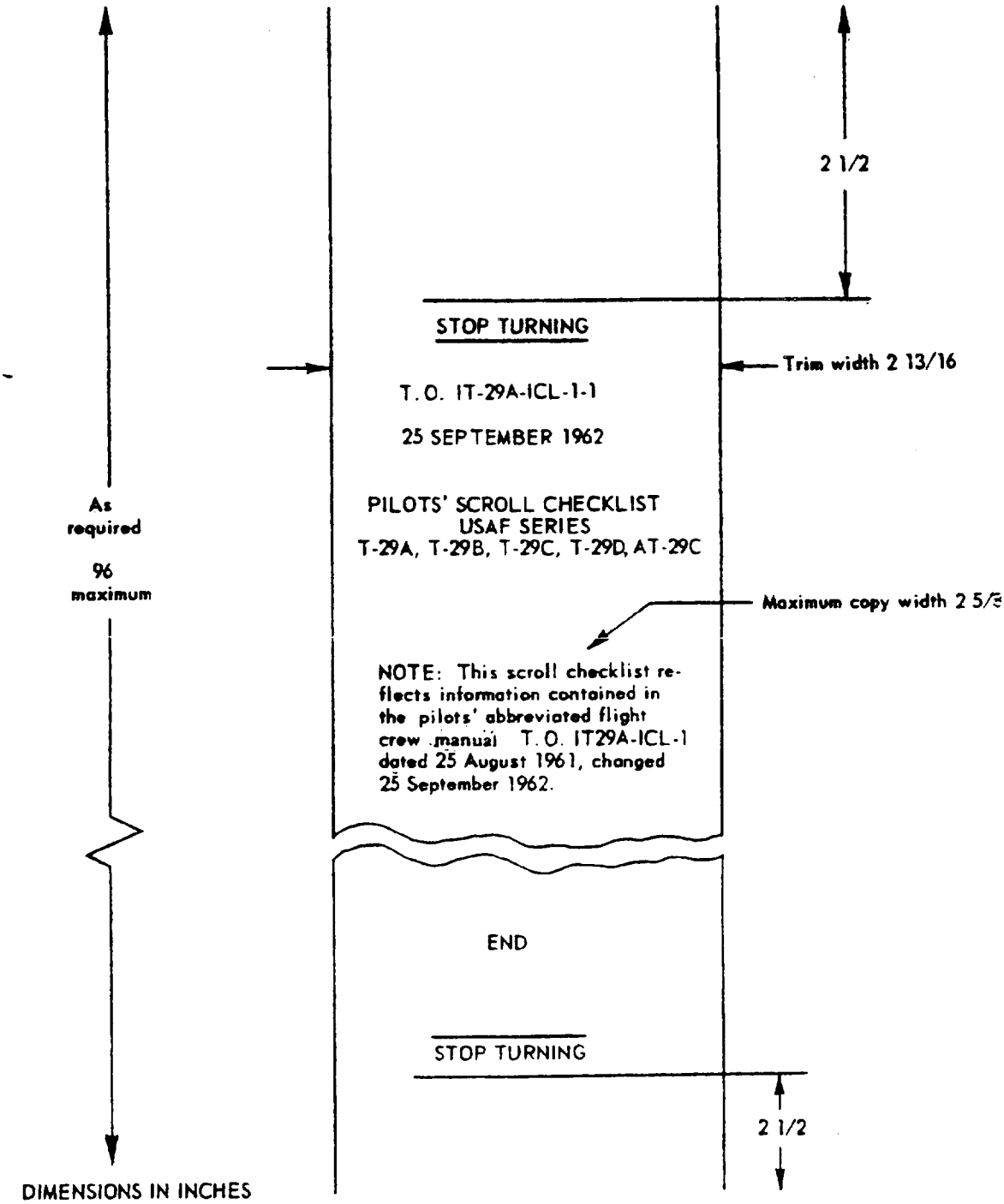


Figure 4. Example of Class 3 (Scroll) Check List

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T.O. 1F-111E-1CL-1

TAKEOFF AND LANDING DATA CARD

CONDITIONS

	Takeoff	Landing
Gross Weight	_____	_____
Flap Setting	_____	_____
Runway Temp	_____	_____
Pressure Altitude	_____	_____
Wind	_____	_____
Runway Length	_____	_____
Runway Slope	_____	_____

TAKEOFF

Engine Pressure	_____Max AB	Sec
Acceleration Check	_____KIAS	_____Feet
Continuation Speed/Dist.	_____KIAS	_____Feet
Takeoff Speed Dist.	_____KIAS	_____Feet
Refusal Speed/Dist.	_____KIAS	_____Feet
Single Engine T/O Speed (300 fpm)	_____KIAS	

LANDING

	Immediately After Takeoff	Final Landing
	G.W._____	G.W._____
Approach Speed	_____	_____
Touchdown Speed	_____	_____
Landing Distance	_____	_____

Figure 5. Example of TOLD Card

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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